

Basic course

KNX RF

KNX Association

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Preface

This training document is created for the Basic course. Sections with a heading marked [E] contain important information that is also required for the NextGen course. This includes any sub-sections within them. If a heading does not have [E], it means the content is only for reference and is not mandatory for the NextGen course.

The documentation focuses on the most used configuration mode in KNX, i.e. S-Mode (System Mode). Configuration modes that used to be supported in the past like E-Mode (Easy Mode) are not covered further.

1 KNX Media [E]

KNX supports multiple communication media, including KNX Twisted Pair (TP), (KNX Powerline), KNX Radio Frequency (RF), KNX Internet Protocol (IP), and KNX Internet of Things (IoT). This flexibility means that KNX is not limited to a single communication medium. Devices can communicate within a specific medium or across multiple media using media couplers. Each communication medium has its own advantages and disadvantages, which are covered in different sections of the KNX training documentation.

This chapter focuses specifically on **KNX Radio Frequency (RF)**.

2 Introduction – KNX RF [E]

KNX RF enables the wireless transmission of telegrams using radio frequency, so there is no need to install additional TP cables. This makes it especially useful when expanding an existing KNX TP installation in places where adding new cables is not possible due to building regulations, aesthetics, or other restrictions—such as in museums or public buildings.

KNX RF devices can be battery-powered, battery-free (e.g., solar-powered), or mains-powered. Battery and energy-harvesting devices have limited transmission power, which affects their range.

To ensure KNX RF systems work correctly, this chapter will cover the basics of radio frequency communication and the KNX RF technology.

3 The RF transmission medium [E]

The range of KNX RF components is measured in an open, unobstructed environment (free-field range). However, the actual range inside a building is much shorter due to obstacles like walls and furniture.

The maximum range in an open space is about 100 meters. This range is limited by regulations on the maximum allowed transmission power and the available energy, especially for battery-powered or battery-free devices (such as those using solar energy).

Inside buildings, radio signals weaken (= signal attenuation) as they travel from the transmitter to the receiver due to various factors. KNX RF operates in the 868 MHz frequency band.

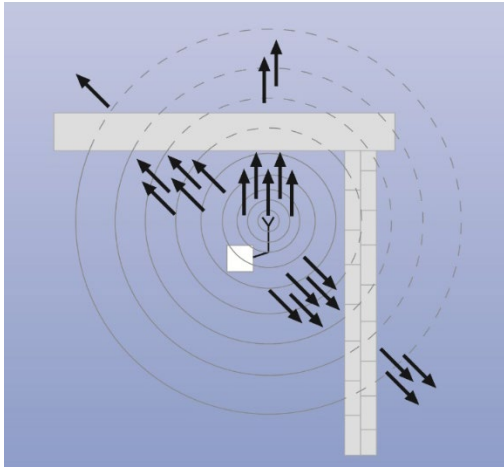


Figure 1: Attenuation by walls and ceilings depending on the material and the wall thickness

Radio signals are weakened when passing through walls, ceilings, and furniture. They don't travel through these obstacles freely but lose strength and can also be partially reflected. Metal objects can either block or reflect radio signals, creating so-called radio shadows. These are areas where the signal cannot be received directly.

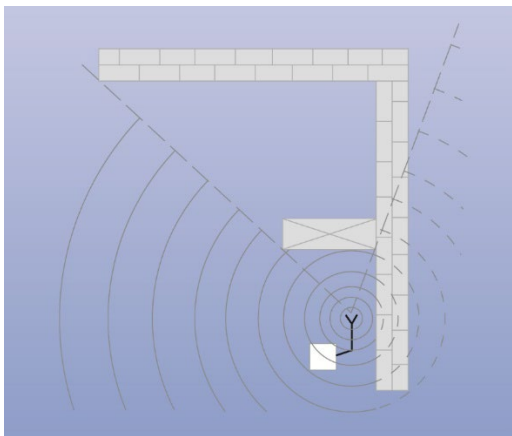


Figure 2: Radio shadows by metal walls, doors or cabinets

Reflections of radio signals can be both positive and negative.

- **Positive effect:** Reflections can help in areas where the direct signal cannot reach, allowing communication to still occur.
- **Negative effect:** If a reflected signal and a direct signal reach the receiver at the same time but with different travel distances (=propagation delay), they may interfere with each other. This can weaken the overall signal and reduce reception quality.

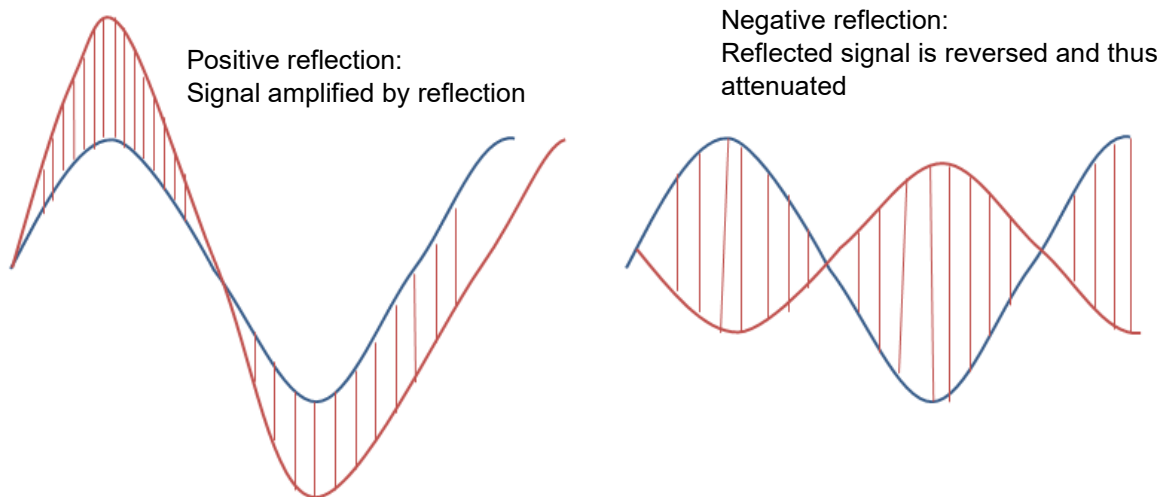


Figure 3: Consequences of reflections

When planning a building, it's important to consider obstacles and other factors that could interfere with radio signals. KNX RF devices should be installed in locations that minimize these issues for the best performance.

If the installation locations are fixed and cannot be freely chosen, a KNX RF repeater can often help improve the signal and ensure a reliable connection. For mains-powered KNX RF devices, make sure that the mains cable is available at the chosen installation location.

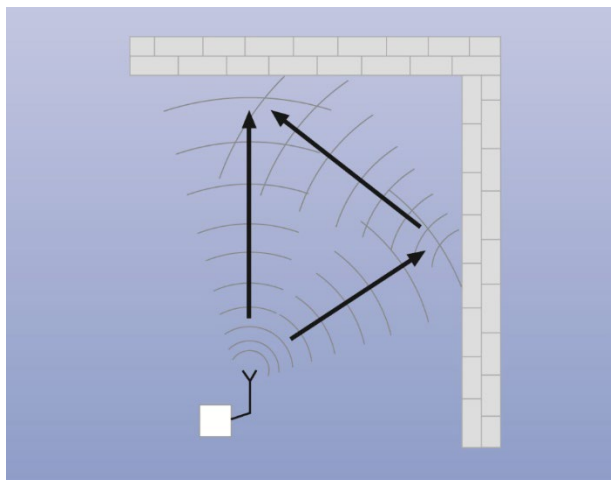


Figure 4: Signal attenuation and amplification by reflections

4 Transmission technology

There are two flavors of the KNX RF technology: the **KNX RF Ready** and **KNX RF Multi** transmission technology.

- **KNX RF Ready**: Uses a **single-channel** communication model, meaning all devices operate on a single fixed frequency.
- **KNX RF Multi**: Uses a **multi-channel** communication model with **frequency hopping**, enhancing reliability and reducing interference.

4.1 KNX RF Ready

In radio technology, information is sent by imposing an input signal onto a **carrier wave**. This modification process is called modulation whereby the shape of the carrier wave is changed so that it can “carry” the information that needs to be sent (e.g. audio, data etc.). Modulation can be done in different ways:

- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)
- Or a combination of these methods.

Once the modulated signal reaches the RF receiver, it goes through a process called demodulation, where the original information is recovered from the signal.

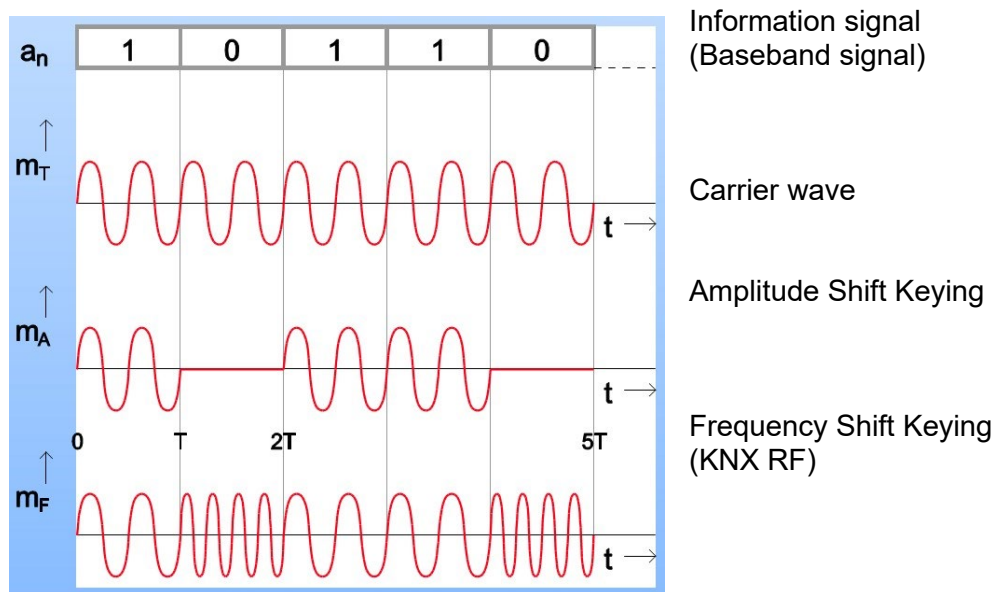


Figure 5: Types of modulation

In the **KNX RF system**, **Frequency Shift Keying (FSK)** is used for sending information.

The basic structure of a transmitter and a receiver is shown in the figure below.

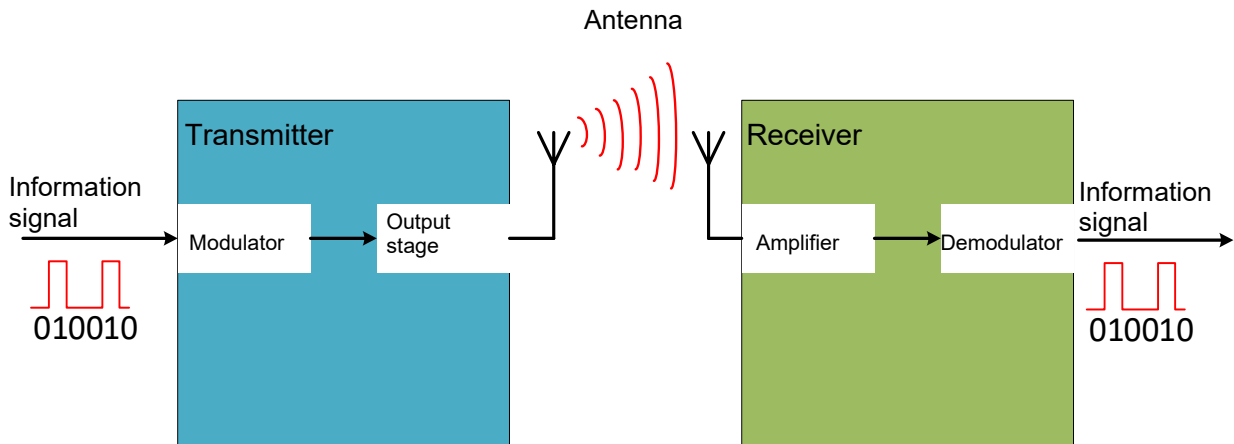


Figure 6: Block diagram of a KNX radio transmission

4.2 KNX RF Multi

In single-channel radio communication (KNX RF Ready), signals can be disrupted by other RF systems operating in the same or nearby frequencies. These disturbances can be caused by:

- Devices that continuously transmit signals.
- Devices with higher transmission power.
- Defective devices.
- Overlapping signals from different sources.

To counter such disruptions, KNX also supports KNX RF-Multi, which can transmit its information over several channels. KNX RF Multi devices are supported starting from ETS6, the latest generation of the KNX configuration software.

4.2.1 How KNX RF Multi works

- Multiple Fast (F) and Slow (S) channels are available for communication.
- If one channel (e.g., F1 or S1) is busy or disturbed, the device can switch (hop) to another channel (e.g., F2, F3 or S2).
- The choice of channels depends on:
 - **Application type:** Fast response needed (e.g., lighting) vs. slow response (e.g., heating/cooling).
 - **Device power source:** Battery-powered devices need to minimize energy consumption (to avoid early depletion of the built-in battery), so they can use different channels than mains-powered devices.

The following table explains which channel types are used for different device types in KNX RF communication. Devices can either transmit (send signals) or receive signals based on their function and power source.

Table 1: Use of Ready/Fast/Slow for different device types

Device Type	Transmitting (Sending data)	Receiving (Getting data)
Mains-powered actuator (e.g., relay switch)	R/F (Ready/Fast channel)	R/F (Ready/Fast channel)
Battery-powered actuator (e.g., heating/cooling valve)	- (Not meaningful)	S (Slow channel)
Battery-powered sensor (e.g., window contact)	R/F (Ready/Fast channel)	- (Not meaningful)
Mains-powered controller	R/F/S (Ready/Fast/Slow channel)	R/F (Ready/Fast channel)

Explanation:

- Mains-powered actuators (e.g., relay switches) can send and receive data on Ready (R) and Fast (F) channels because they don't have power limitations.
- Battery-powered actuators (e.g., heating/cooling valves) only receive data on Slow (S) channels to save battery life.
- Battery-powered sensors (e.g., window contacts) only send data on Ready (R) and Fast (F) channels, as they don't need to receive information.
- Mains-powered controllers can send and receive on all channel types (R/F/S) since they have a constant power supply, but they can only receive data on Ready (R) and Fast (F) channels.

This setup ensures **efficient power use** for battery-operated devices while allowing fast and reliable communication for mains-powered components.

KNX RF uses **fast** and **slow** channels for communication, each with different data rates and transmission rules.

Data transmission speed:

- Fast channels send data at 16,384 kbps.
- Slow channels send data at half that speed (mainly due to the longer preamble needed for synchronization).

Transmission power and duty cycle¹:

The **duty cycle** depends on the channel and transmission power:

- Fast Channels (F1 & F2):
 - The duty cycle is 1% or 0.1% when the maximum transmission power is 25 mW.
- Fast Channel (F3) & Slow Channel (S1):
 - Duty cycle can be 100% when the maximum transmission power is 5 mW.
 - If transmission power is between 5 mW and 25 mW, the duty cycle is limited to 1%.
- Slow Channel (S2):
 - Duty cycle is 10% when the maximum transmission power is 25 mW.

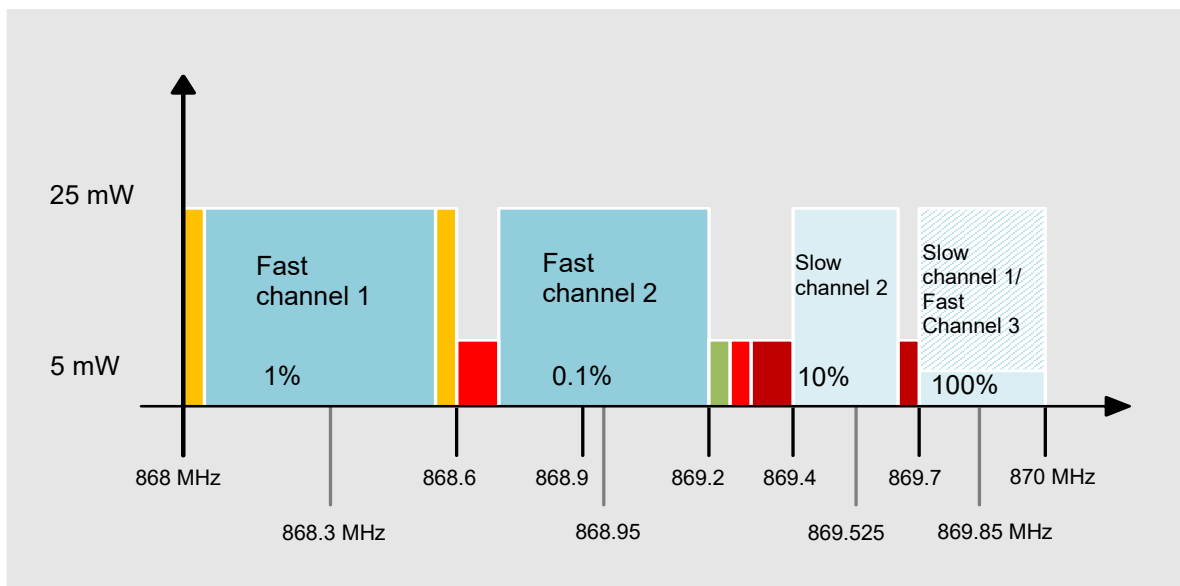


Figure 7: KNX RF-Multi different channels and allocated frequencies

To save energy, KNX RF devices enter sleep mode when not in use. This reduces power consumption by:

- Up to 80% for fast channels
- Up to 99% for slow channels

¹ Duty cycle means "The fraction of time the transmitter is actively sending signals compared to the total time available, expressed as a percentage". For a 1% duty cycle, if we look at 1-hour (3600 seconds) window, this means 1% of 3600 s = 36 seconds. That means the transmitter can only be actively transmitting for 36 seconds per hour.

Since devices need to receive telegrams, they periodically wake up to check for incoming telegrams.

Compatibility between KNX RF Ready and KNX RF Multi devices:

A compatibility scheme ensures that KNX RF Ready devices can work with KNX RF Multi devices:

- New KNX RF Ready devices must use a longer preamble to maintain compatibility with KNX RF Multi devices.
- KNX RF Multi devices should be able to switch to KNX RF Ready mode when needed.
- KNX RF Ready devices have the “**KNX RF1.R**” label whereas KNX RF Multi devices have the label “**KNX RF1.M**”.

A KNX RF Multi sender operating on a slow channel uses a preamble of 500 ms instead of the usual 15 ms (compared to 5 ms for KNX RF Ready). This extended preamble allows receivers to remain in sleep mode longer, thereby conserving battery life.

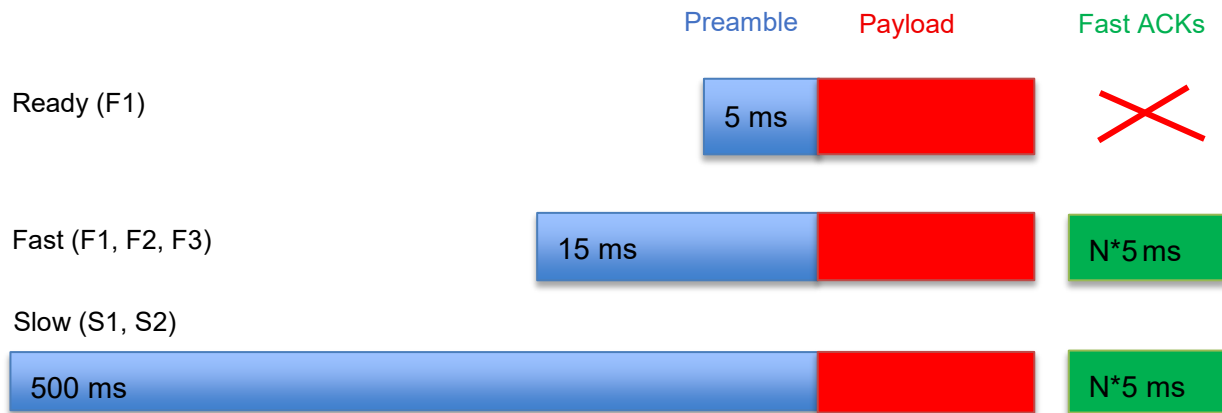


Figure 8: Preamble and different KNX RF flavors

Not to scale

The multi-channel approach in KNX RF improves the chances of successful communication by allowing messages to be sent over different channels. Additionally, it ensures correct reception through a fast acknowledgment system.

Fast Immediate Acknowledgment (Fast IACK):

- Up to 64 devices can send instant confirmations (Fast IACK) after receiving a telegram.
- If a device does not send an acknowledgment within one of the 5 ms time slots (see above), the telegram is automatically resent.
- KNX RF Repeaters help forward these confirmations, ensuring messages are successfully received even over long distances.

Collision Avoidance:

- Before sending a telegram, KNX RF checks if another transmission is already ongoing.
- This reduces signal collisions and improves overall communication efficiency.

Security features & Long Frames:

- KNX RF Multi must support KNX Secure. This means that it also supports long frames which are required for the KNX Secure Application Layer.
- Security features include:
 - Authentication (verifying sender identity).
 - Encryption (protecting message content).
 - Sequence number (each telegram includes a unique sequence number)
- These security measures ensure that media couplers (which connect different KNX communication media) are not weak points in the system.

Suitable for critical applications:

Because of its enhanced security and reliability, KNX RF can be used in applications such as:

- Metering
- Access control
- Fire alarm systems
- ...

4.3 Bus access in KNX RF

In KNX RF, devices send telegrams using two different approaches based on whether they are **unidirectional** or **bidirectional**.

4.3.1 Unidirectional devices (One-way communication)

- These devices send messages immediately when needed and a second one randomly after the first.
- Since the duty cycle is only 1%, meaning devices can only transmit for short periods, collisions are rare.

4.3.2 Bidirectional devices (Two-way communication)

- Before sending a message, these devices check if the radio channel is free.
- If the channel is busy, they wait until it becomes free before transmitting.
- This method prevents collisions and ensures more reliable communication.

4.3.3 Semi-bidirectional devices

- To extend the lifetime of the included battery, some sensors are semi-bidirectional, meaning, they are able to send and receive messages when being downloaded by the ETS, but behave unidirectional when not being in programming mode.

By using these strategies, KNX RF ensures efficient and reliable data transmission while minimizing the risk of communication failures.

4.4 Structure of KNX RF devices

KNX RF devices are available in different installation types:

1. Flush-mounted devices – Installed inside walls or electrical boxes, usually for switching lights, dimming, controlling shutters, or mains-powered equipment. These flush-mounted devices behave like actuators. In order to operate their function, push buttons are often clipped onto them.
2. Surface-mounted or built-in devices – Include sensors, actuators, or a combination of the latter devices that can be mounted, glued, or embedded into various surfaces.

Devices can be unidirectional or bidirectional depending on their function and application.

Unidirectional devices:

- These are typically sensors with limited energy source. To conserve energy, they only transmit telegrams when triggered by events.
- Typically, battery-powered sensors or detectors such as:
 - Handheld or wall-mounted transmitters
 - Binary inputs
 - Door/window contacts

Bidirectional devices:

- These can both send and receive messages.
- These devices often act as both sensors and actuators.
- KNX RF repeaters are also bidirectional devices helping to extend the wireless range by automatically forwarding RF messages.
- Some bidirectional actuators (e.g., relays) include a built-in repeater function.

Media couplers are used to connect KNX RF to KNX TP.

5 Topology

5.1 No hierarchical structure

- KNX RF devices do not follow a strict hierarchy - they can be installed anywhere within range.
- If they are within radio range, any sensor can communicate directly with any actuator.

5.2 Radio range & interference prevention

- The exact radio range cannot be precisely defined because signals may reach devices in neighboring KNX RF installations.
- To prevent interference between different KNX RF systems:
 - KNX RF devices use domain addresses.
 - Only receivers that are linked to the transmitter in the same domain will process the transmitted telegrams.

As explained in the first paragraphs, radio signals can be weakened by walls, ceilings, and furniture, reducing the effective range inside buildings. The range can be extended by using:

- Up to **two** KNX RF repeaters for **KNX RF Multi devices** or
- Up to **six** KNX RF repeaters for **KNX RF Ready devices**.

Repeaters help transmit signals over multiple floors and ensure communication reaches all necessary devices.

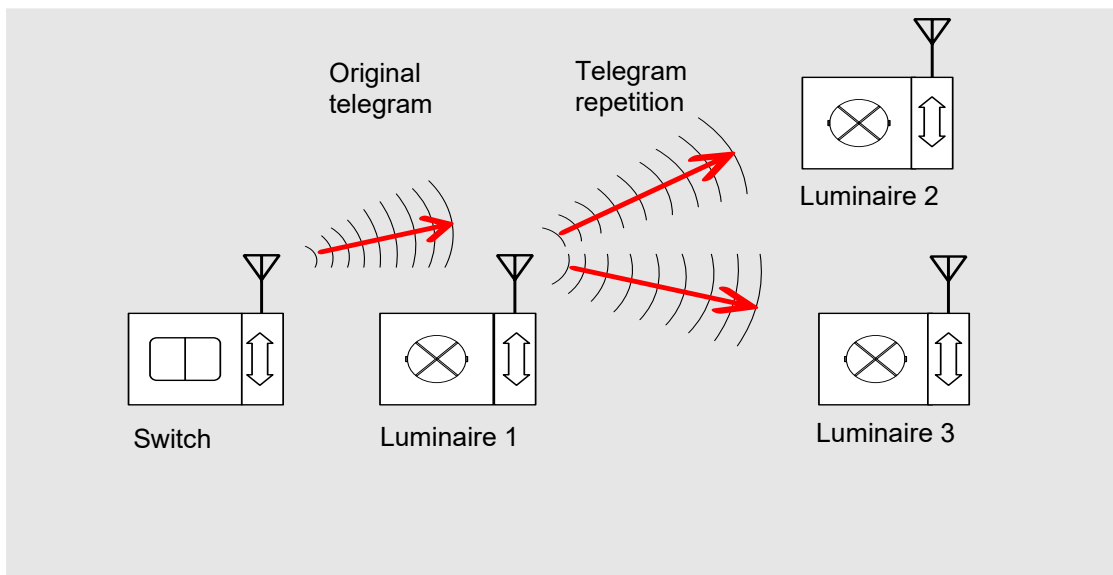


Figure 9: Repeater function in KNX RF: Luminaire 1 increases the range of the switch through telegram repetition.

5.3 Combining different transmission media in KNX

One of the biggest advantages of **KNX** is that it supports multiple **transmission media** within the same system. These include:

- **KNX Twisted Pair (TP)** – Wired communication.
- **KNX Radio Frequency (RF)** – Wireless communication.
- **KNX IP** – Communication over Ethernet/IP networks.
- **KNX IoT** – Communication over IPv6 (wireless and wired communication)

5.3.1 Mixing KNX RF with other media

A KNX system can be:

- Fully wireless, using only KNX RF devices.
- A hybrid system, combining KNX RF with KNX TP (wired devices).

A media coupler is used to transfer telegrams between devices on different communication media.

5.3.2 Media couplers and repeaters

- Media couplers act as a bridge between KNX RF and KNX TP devices.
- Some mains-powered devices can also function as repeaters to extend the wireless range.
- The media coupler (which connects a wired bus line to an RF line²) works like a line coupler – or since the introduction of ETS6 – also like a segment coupler. See the appropriate paragraph in the “Topology” chapter of the Basic course.

² For Segment coupler, please consult the appropriate clause in the section ‘Topology’.

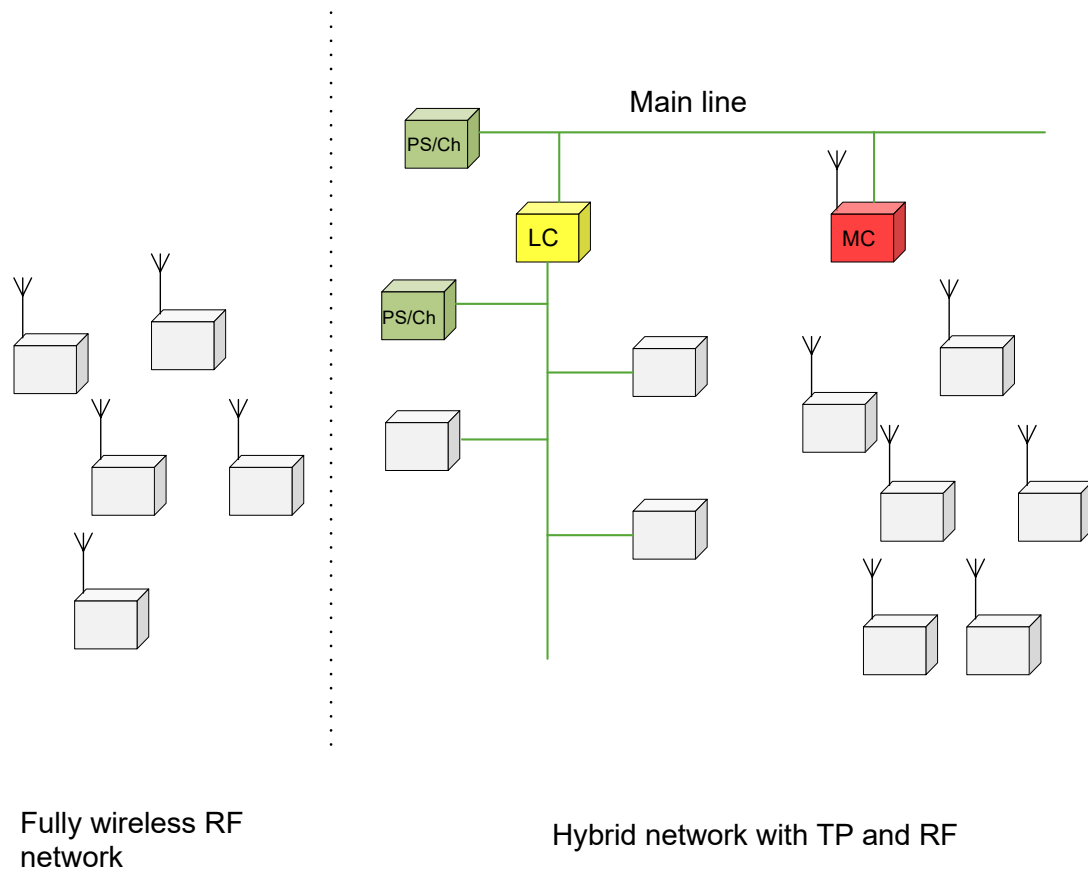


Figure 10: Topology of RF

The above picture shows an RF topology, where the media coupler is installed in the main line. Newer media couplers also provide segment coupler functionality, not obliging the mounting in the main line, but directly in the line.

6 Commissioning and system checks in KNX RF

6.1 Checking the installation (safety and compliance)

- After installing a KNX RF system, standard low-voltage safety tests must be performed.
- The installation must follow the regulations for mains power systems up to 1,000 V to ensure compliance and safety.

6.2 Commissioning and programming KNX RF devices

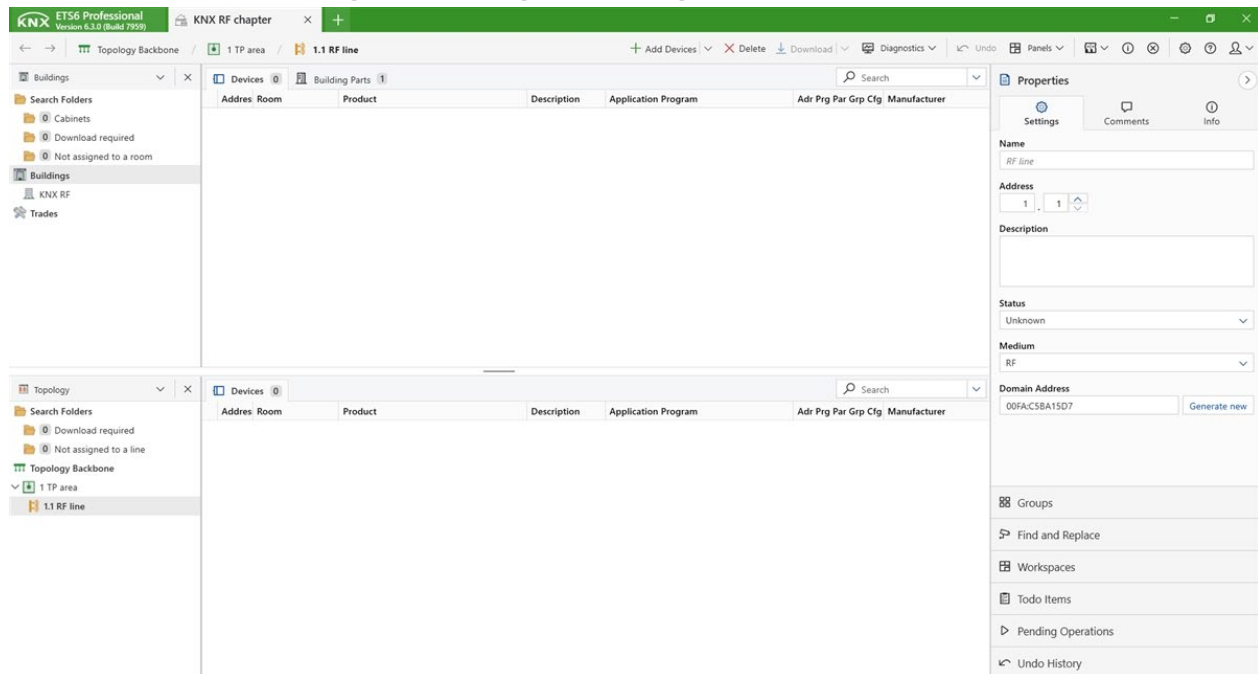


Figure 11: Domain address in ETS6

The commissioning process for KNX RF devices is very similar to that of KNX TP devices, with a few key differences related to wireless communication:

Unique domain address for each RF line:

- Every KNX RF line or segment is assigned a unique domain address.
- This prevents interference between different KNX RF installations located close to each other.
- Devices within the same domain address can communicate, while others are ignored.

Configuration using a KNX RF data interface:

- A KNX RF data interface allows direct and wireless configuration and commissioning of KNX RF devices from a PC using the ETS software.



Figure 12: KNX RF data interface

Much of the complexity in combining KNX RF (Ready, Fast, Slow) devices in ETS is hidden from the ETS user. When configuring a KNX RF installation, ETS automatically adjusts settings to ensure devices communicate properly:

Matching a sender to a slow receiver:

- If a device can send messages in Ready, Fast, and Slow, but it is linked to a receiver that only supports Slow, ETS ensures the sender transmits in Slow mode.

Matching a Ready sender to a Fast receiver:

- If a receiver supports both Ready and Fast, but the sender only supports Ready, ETS configures the sender to send the information on Ready.

Handling multiple devices with different RF types:

If a device needs to send data to three different devices and each supports a different RF type (Ready, Fast, Slow), ETS configures the sender to send the same payload in three different telegrams:

- **Once in Ready mode** for the Ready receiver.
- **Once in Fast mode** for the Fast receiver.
- **Once in Slow mode** for the Slow receiver.

ETS includes a capability matrix, which allows it to automatically determine how different KNX RF devices should communicate.

Table 2: Capability matrix for KNX RF in ETS

Sending Capabilities	Receiving Capabilities		
	R	R+F	S
R	R	R	
R+F	R	F	
R+F+S	R	F	S

How ETS uses this matrix:

- If a sender only supports Ready (R), ETS ensures it only communicates with receivers that support R.
- If a sender supports Ready and Fast (R+F), ETS allows it to communicate using R with R-only receivers, and F with F-capable receivers.
- If a sender supports Ready, Fast, and Slow (R+F+S), ETS ensures it sends R to R-receivers, F to F-receivers, and S to S-receivers.